

Indian Institute Of Technology Gandhinagar



Assignment Report

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Two-Stage OPAMP Design

Design a two-stage OPAMP with a Folded Cascode first stage, followed by a Common Source (CS) stage (refer to Fig. 1) for the following specifications. Use the previously designed Voltage and Current Reference for biasing. Miller Compensation must be used to achieve the desired circuit performance.

Design Specifications

- Differential Gain > 100 dB
- Common Mode Rejection Ratio (CMRR) > 100 dB
- Output Swing > 1 V
- Unity Gain Bandwidth (UGB) > 10 MHz
- Phase Margin (PM) $> 65^\circ$
- Power Dissipation $< 1mW$
- CL = 1 pF

PMOS W/L Calculation

The drain current in saturation is given by:

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} V_{ov}^2$$

Rearranging:

$$\frac{W}{L} = \frac{2I_D}{\mu_p C_{ox} V_{ov}^2}$$

Given:

$$I_D = 6.75 \mu A, \quad \mu_p C_{ox} = 65 \mu A/V^2, \quad V_{ov} = 0.4 V$$

Substituting:

$$\begin{aligned} \frac{W}{L} &= \frac{2 \times 6.75 \times 10^{-6}}{65 \times 10^{-6} \times (0.4)^2} \\ &= \frac{13.5 \times 10^{-6}}{65 \times 10^{-6} \times 0.16} \\ &= \frac{13.5}{10.4} \approx 1.30 \end{aligned}$$

Thus,

$$\frac{W}{L} \approx 1.30$$

Folded PMOS W/L Calculation

The drain current in saturation is given by:

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} V_{ov}^2$$

Rearranging:

$$\frac{W}{L} = \frac{2I_D}{\mu_p C_{ox} V_{ov}^2}$$

Given:

$$I_D = 2.25 \mu A, \quad \mu_p C_{ox} = 65 \mu A/V^2, \quad V_{ov} = 0.24 V$$

Substituting:

$$\begin{aligned} \frac{W}{L} &= \frac{2 \times 2.25 \times 10^{-6}}{65 \times 10^{-6} \times (0.24)^2} \\ &= \frac{4.5 \times 10^{-6}}{65 \times 10^{-6} \times 0.0576} \\ &= \frac{4.5}{3.744} \approx 1.20 \end{aligned}$$

Thus,

$$\frac{W}{L} \approx 1.20$$

Folded NMOS W/L Calculation

The drain current in saturation is given by:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

Rearranging:

$$\frac{W}{L} = \frac{2I_D}{\mu_n C_{ox} V_{ov}^2}$$

Given:

$$I_D = 2.25 \mu A, \quad \mu_n C_{ox} = 300 \mu A/V^2, \quad V_{ov} = 0.18 V$$

Substituting:

$$\begin{aligned}\frac{W}{L} &= \frac{2 \times 2.25 \times 10^{-6}}{300 \times 10^{-6} \times (0.18)^2} \\ &= \frac{4.5 \times 10^{-6}}{300 \times 10^{-6} \times 0.0324} \\ &= \frac{4.5}{9.72} \approx 0.46\end{aligned}$$

Thus,

$$\frac{W}{L} \approx 0.46$$

Transconductance Calculation

The transconductance is given by:

$$g_m = \frac{2I_D}{V_{ov}}$$

Given:

$$I_D = 4.5 \mu A, \quad V_{ov} = 0.18 V$$

Substituting:

$$\begin{aligned}g_m &= \frac{2 \times 4.5 \times 10^{-6}}{0.18} \\ &= \frac{9 \times 10^{-6}}{0.18} = 50 \times 10^{-6}\end{aligned}$$

Thus,

$$g_m = 50 \mu S$$

W/L Calculation from g_m

The transconductance relation is:

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

Rearranging:

$$\frac{W}{L} = \frac{g_m^2}{2\mu_n C_{ox} I_D}$$

Given:

$$g_m = 50 \mu S, \quad \mu_n C_{ox} = 300 \mu A/V^2, \quad I_D = 4.5 \mu A$$

Substituting:

$$\begin{aligned}\frac{W}{L} &= \frac{(50 \times 10^{-6})^2}{2 \times 300 \times 10^{-6} \times 4.5 \times 10^{-6}} \\ &= \frac{2.5 \times 10^{-9}}{2.7 \times 10^{-9}} \approx 0.93\end{aligned}$$

Thus,

$$\frac{W}{L} \approx 0.93$$

Tail NMOS W/L Calculation

The drain current in saturation is given by:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

Rearranging:

$$\frac{W}{L} = \frac{2I_D}{\mu_n C_{ox} V_{ov}^2}$$

Given:

$$I_D = 9 \mu A, \quad \mu_n C_{ox} = 300 \mu A/V^2, \quad V_{ov} = 0.18 V$$

Substituting:

$$\begin{aligned}\frac{W}{L} &= \frac{2 \times 9 \times 10^{-6}}{300 \times 10^{-6} \times (0.18)^2} \\ &= \frac{18 \times 10^{-6}}{300 \times 10^{-6} \times 0.0324} \\ &= \frac{18}{9.72} \approx 1.85\end{aligned}$$

Thus,

$$\frac{W}{L} \approx 1.85$$

Common Source Stage Design

Current Calculation:

$$g_m = \frac{2I_D}{V_{ov}} \Rightarrow I_D = \frac{g_m V_{ov}}{2}$$

$$I_D = \frac{100 \times 10^{-6} \times 0.18}{2} = 9 \mu A$$

PMOS W/L:

$$\begin{aligned} \frac{W}{L} &= \frac{2I_D}{\mu_p C_{ox} V_{ov}^2} \\ &= \frac{18 \times 10^{-6}}{65 \times 10^{-6} \times (0.18)^2} = \frac{18}{2.106} \approx 8.55 \end{aligned}$$

$$\frac{W}{L}_{PMOS} \approx 8.55$$

NMOS W/L:

$$\begin{aligned} \frac{W}{L} &= \frac{2I_D}{\mu_n C_{ox} V_{ov}^2} \\ &= \frac{18}{300 \times 0.0324} = \frac{18}{9.72} \approx 1.85 \end{aligned}$$

$$\frac{W}{L}_{NMOS} \approx 1.85$$

Third Stage (Common Drain) W/L Calculation with Higher Current

The drain current in saturation is given by:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

Rearranging:

$$\frac{W}{L} = \frac{2I_D}{\mu_n C_{ox} V_{ov}^2}$$

Given:

$$I_D = 50 \mu A, \quad \mu_n C_{ox} = 300 \mu A/V^2, \quad V_{ov} = 0.18 V$$

Substituting:

$$\begin{aligned} \frac{W}{L} &= \frac{2 \times 50 \times 10^{-6}}{300 \times 10^{-6} \times (0.18)^2} \\ &= \frac{100 \times 10^{-6}}{300 \times 10^{-6} \times 0.0324} \end{aligned}$$

$$= \frac{100}{9.72} \approx 10.3$$

Thus,

$$\frac{W}{L} \approx 10.3$$

Transistor	Type	W (μm)	L (μm)	V_{ov} (V)
M1	NMOS (Input)	0.925	1	0.088
M2	NMOS (Input)	0.925	1	0.088
M3	PMOS (up)	1.31	1	0.23
M4	PMOS (up)	1.31	1	0.23
M5	NMOS (Tail)	1.85	1	0.084
M6	NMOS (Folded)	0.465	1	0.145
M7	PMOS (Load)	1.21	1	0.177
M8	PMOS (Load)	1.21	1	0.177
M9	NMOS (2nd Stage)	1.65	1	0.384
M10	PMOS (2nd Stage Load)	8.545	1	0.377
M11	NMOS (3rd Stage - Common Drain)	10.5	1	0.1

Table 1: Transistor Sizing Summary

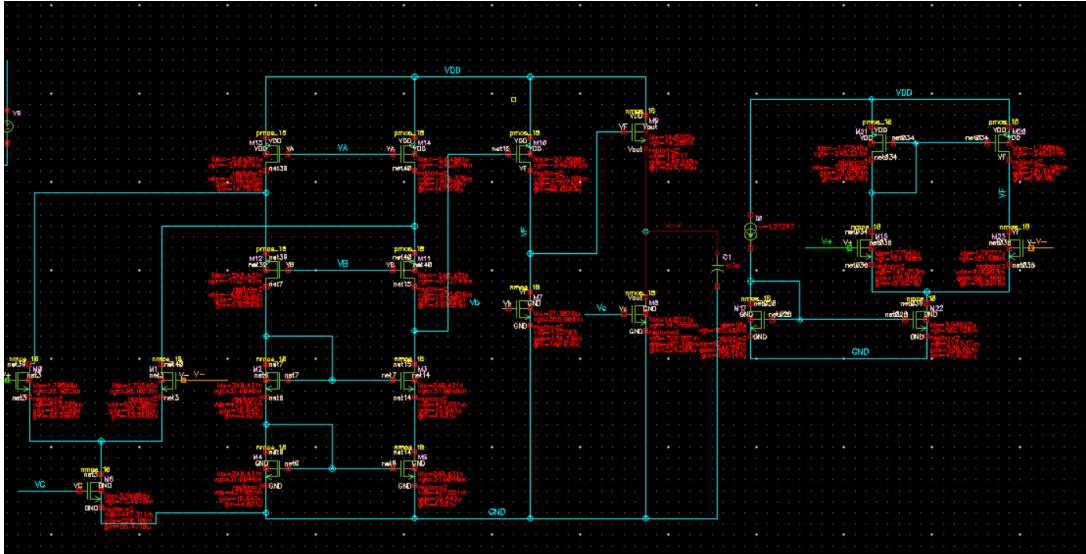


Figure 1: Circuit Diagram for Folded cascode opamp with miller compensation

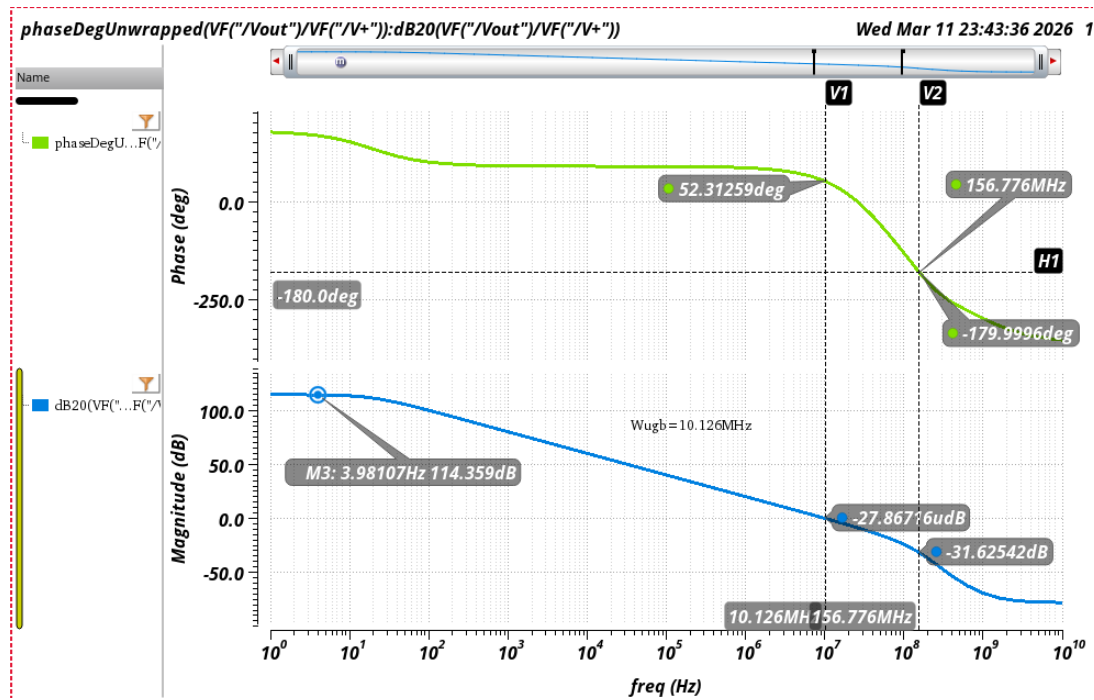


Figure 2: Gain plot for Folded opamp with miller compensation

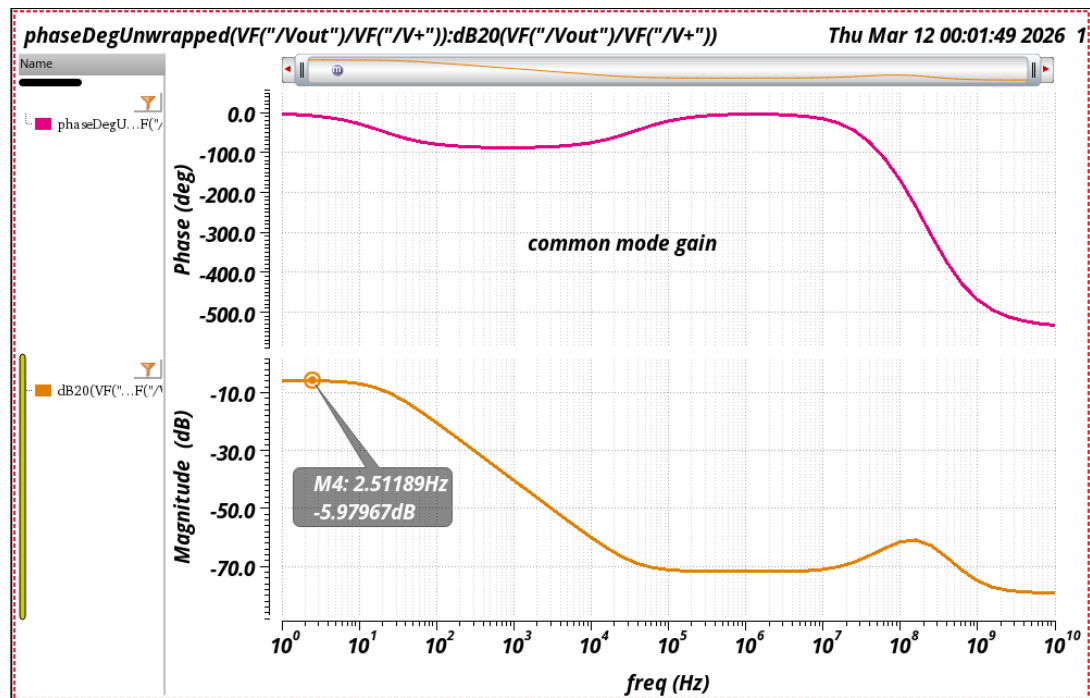


Figure 3: Common Mode GAIN plot for Folded opamp with miller compensation

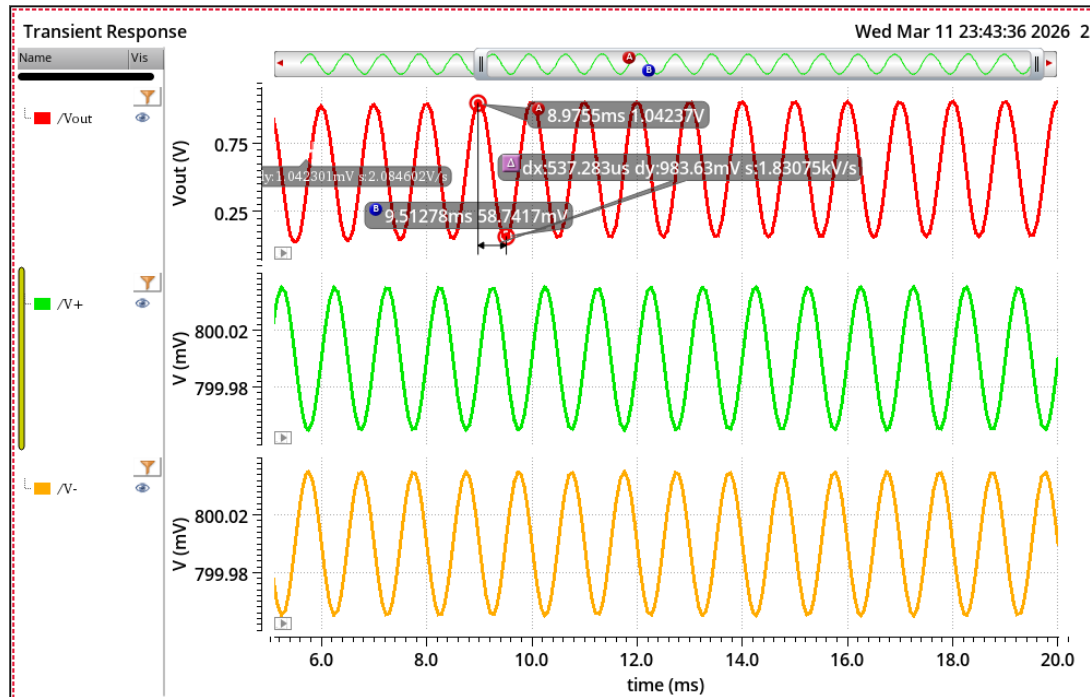


Figure 4: Swing

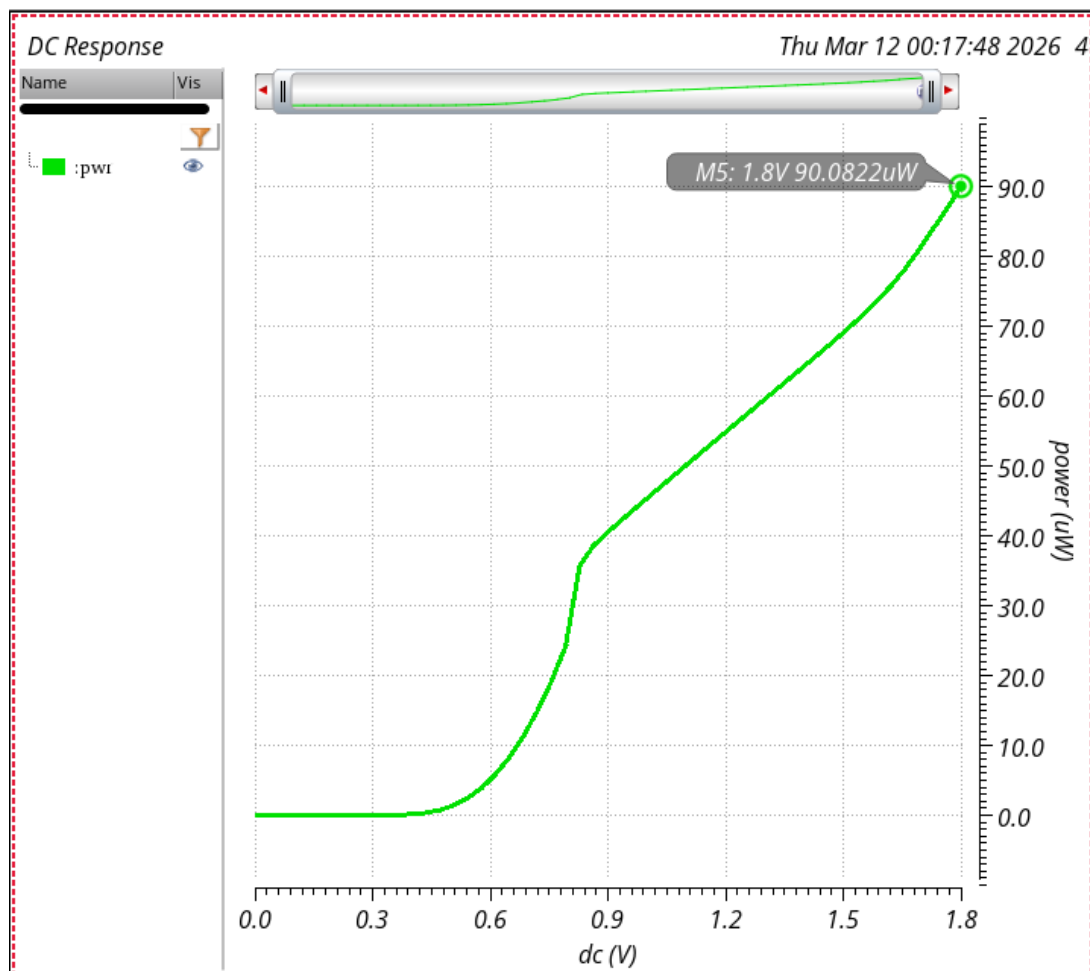


Figure 5: Power Consumption

Process Corner	Gain(dB)	CMRR(dB)	UGB(MHz)	PM(deg)	Output Swing(V)
TT	114.359	120.338	10.126	52.3125	0.983
FF	122.56	129	12.367	51.88	1.15
SS	105.9	123.2	6.996	56.06	1.064
SF	119.4	114.2	9.084	53.7	1.074
FS	126.087	122.08	11.23	49.02	1.095

Table 2: Performance of Designed folded OPAMP Across Process Corners

TASK 2: Designing a Two-Stage OpAmp using Feed-forward Compensation [6 pts]

Objective

Design a two-stage operational amplifier consisting of a Folded Cascode first stage followed by a Common Source (CS) second stage, as shown in Fig. 2. The design should utilize the previously developed voltage and current reference circuits for biasing. Feedforward compensation must be employed to achieve the desired performance.

Specifications

The designed OpAmp must satisfy the following requirements:

- Differential Gain $A_v > 100$ dB
- Common Mode Rejection Ratio (CMRR) > 100 dB
- Output Swing > 1 V
- Unity Gain Bandwidth (UGB) > 50 MHz
- Phase Margin (PM) $> 55^\circ$
- Power Dissipation < 1 mW
- Load Capacitance $C_L = 1$ pF

Transistor	Type	W (μm)	L (μm)	V_{ov} (V)
M1	NMOS (Input)	3	1	0.028
M2	NMOS (Input)	3	1	0.028
M3	PMOS (up)	1.31	1	0.246
M4	PMOS (up)	1.31	1	0.246
M5	NMOS (Tail)	1.85	1	0.189
M6	NMOS (Folded)	0.465	1	0.192
M7	PMOS (Load)	1.21	1	0.149
M8	PMOS (Load)	1.21	1	0.149
M9	NMOS (2nd Stage)	1.65	1	0.365
M10	PMOS (2nd Stage Load)	8.545	1	0.546
M11	NMOS (3rd Stage - Common Drain)	10.5	1	0.126
M13		0.75	1	0.151
M12		0.575	1	0.576

Table 3: Transistor Sizing Summary

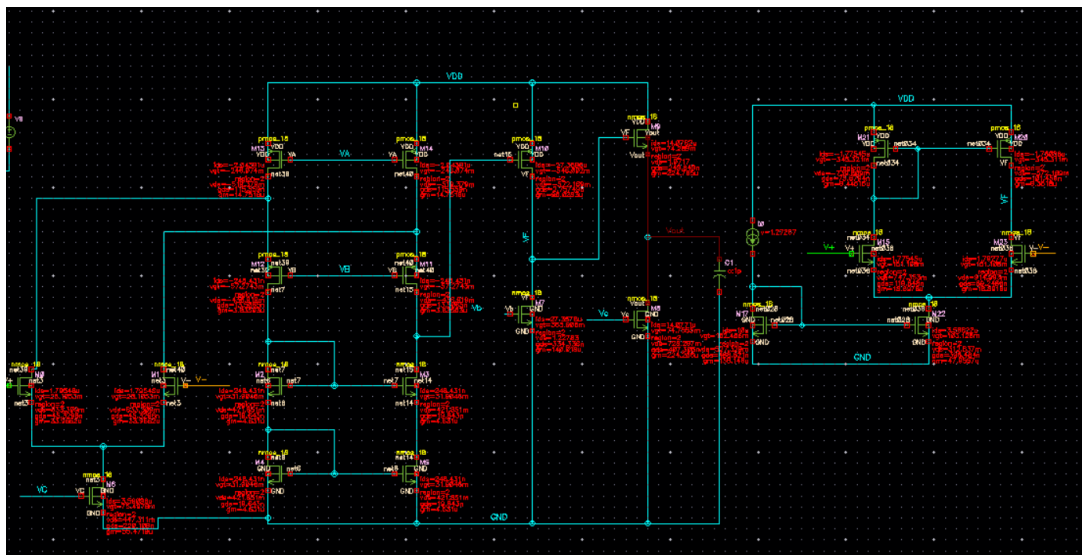


Figure 6: Schematic of feed forward compensation

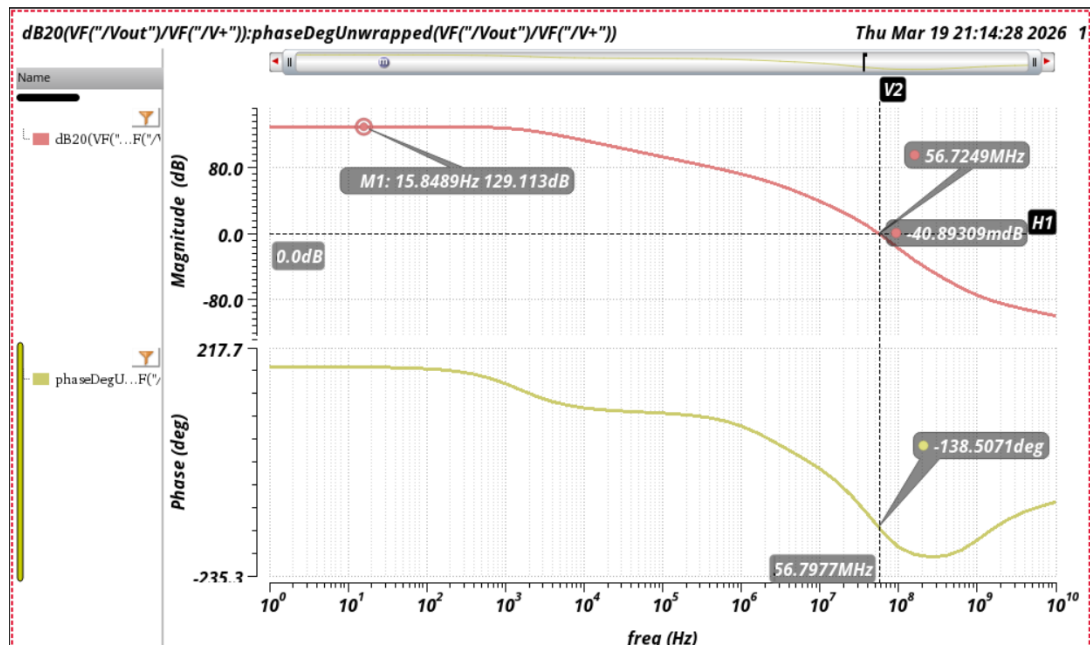


Figure 7: Magnitude plot for differential gain for feed forward compensation

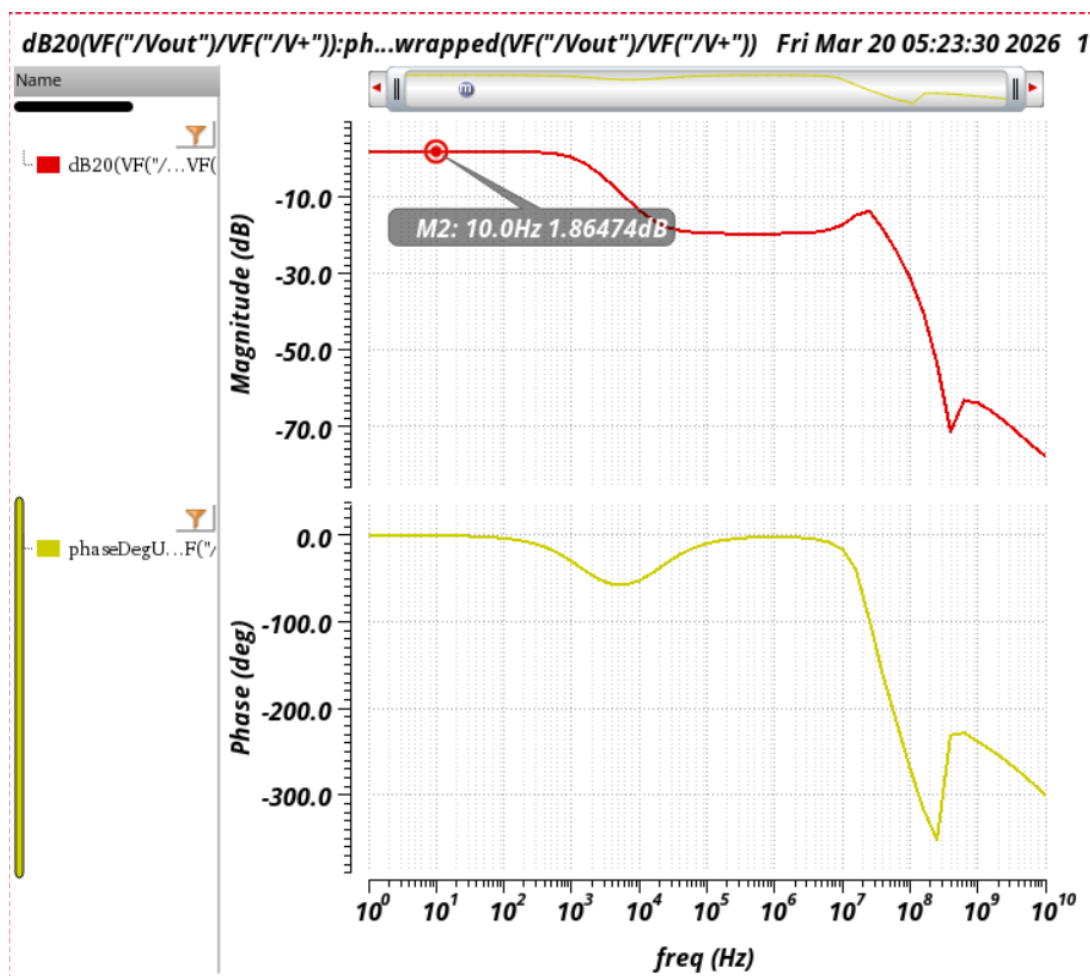


Figure 8: Magnitude plot for Common gain for Feed forward compensation

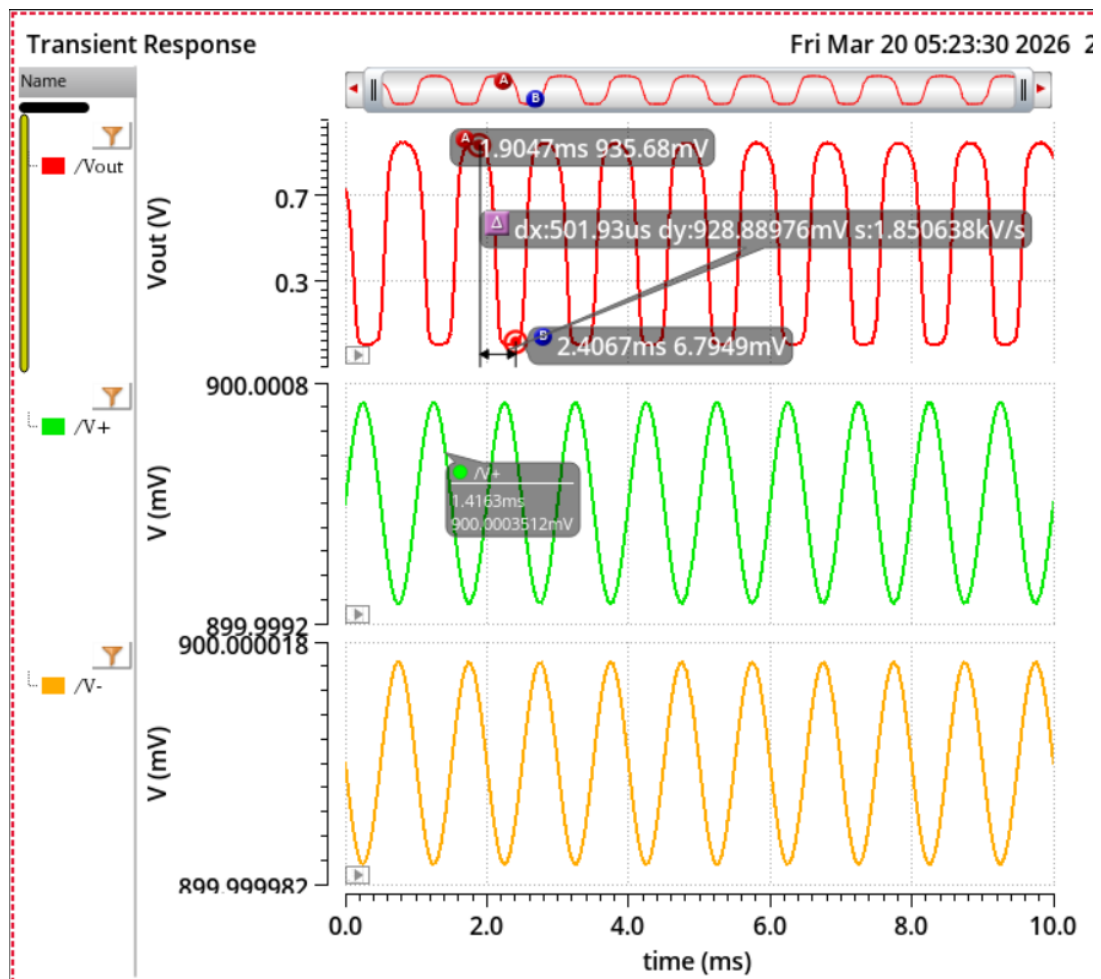


Figure 9: Outout Swing for feed forward compensation

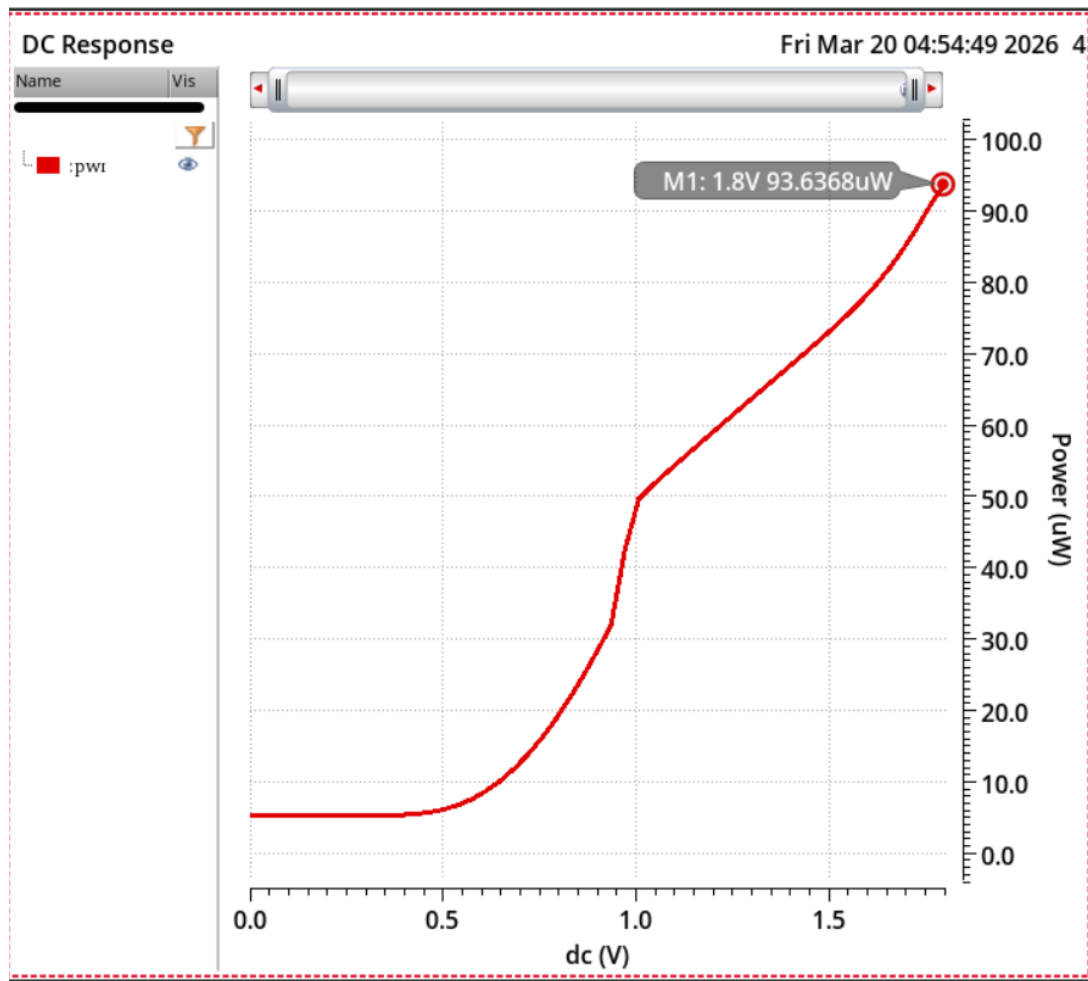


Figure 10: Power Consumption for feed forward compensation

Table 4: Feedforward Compensation Performance Across Process Corners

Process Corner	Gain (dB)	CMRR (dB)	UGB (MHz)	Output Swing (V)
TT	129.31	127.2	52.3125	0.928
FF	124.3	122.6	52.17	0.91
SS	115.9	109.5	46.8	1.023
SF	119.4	112.3	51.2	0.83
FS	122.8	106	49.02	1.06

TASK 3: Designing a Two-Stage OpAmp using Nested-Miller Compensation [4pts]

1. Design a 2-stage OPAMP with a Folded Cascode first stage, followed by a CS stage (refer to Fig. 3) for the following specifications. Use the previously designed Voltage and Current Reference for biasing. You need to use Nested Miller Compensation to design the desired circuit for this task.

- Differential Gain > 140 dB
- CMRR > 140 dB
- Output Swing > 1 V
- UGB > 5 MHz
- PM $> 65^\circ$
- Power dissipation < 1 mW
- $C_L = 1$ pF

Transistor	Type	W (μm)	L (μm)	V_{ov} (V)
M1	NMOS (Input)	0.79	1	0.157
M2	NMOS (Input)	0.79	1	0.157
M3	PMOS (Current Mirror Load)	0.575	1	0.517
M4	PMOS (Current Mirror Load)	0.575	1	0.5117
M5	NMOS (Tail Current Source)	1.235	1	0.174

Table 5: 5T OTA Transistor Design Parameters

the value of miller cap used is 300f F and this is calculated from the Wugb requirement and the folded cascode used is same as the task 1

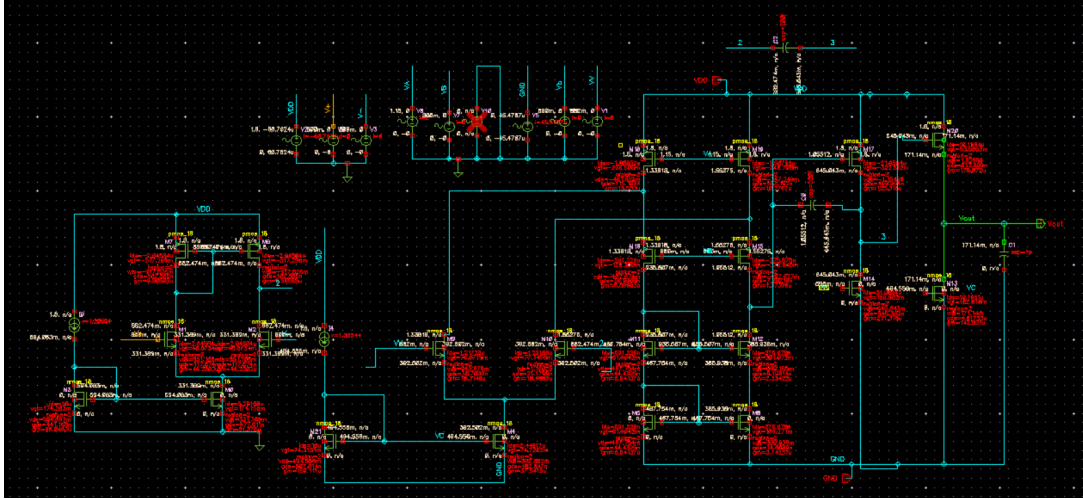


Figure 11: Circuit for differential gain for nested miller compensation

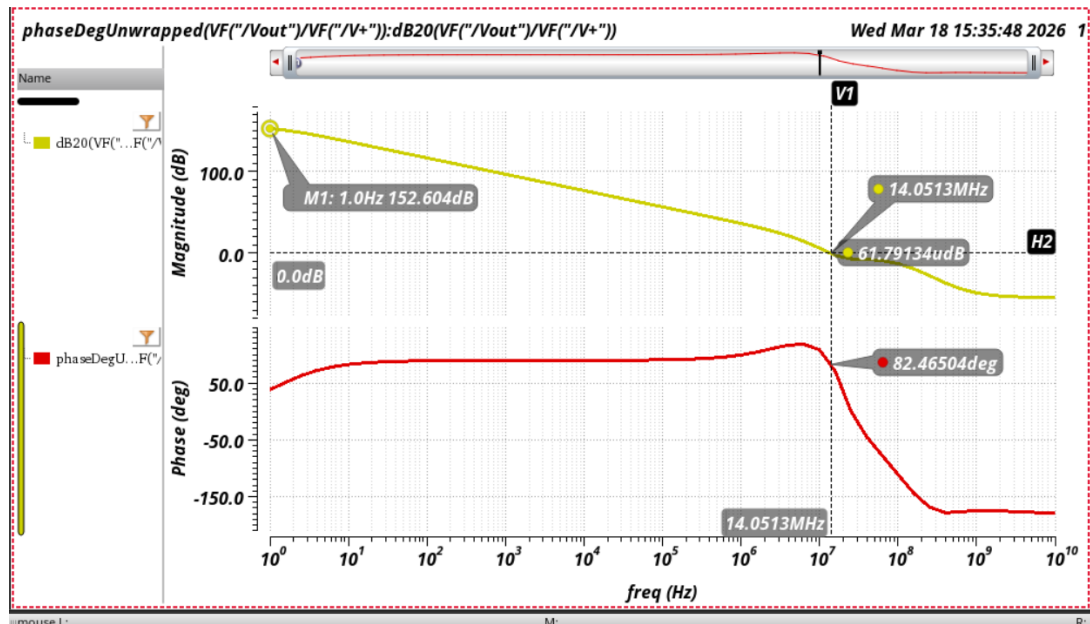


Figure 12: Magnitude plot for differential gain for nested miller compensation

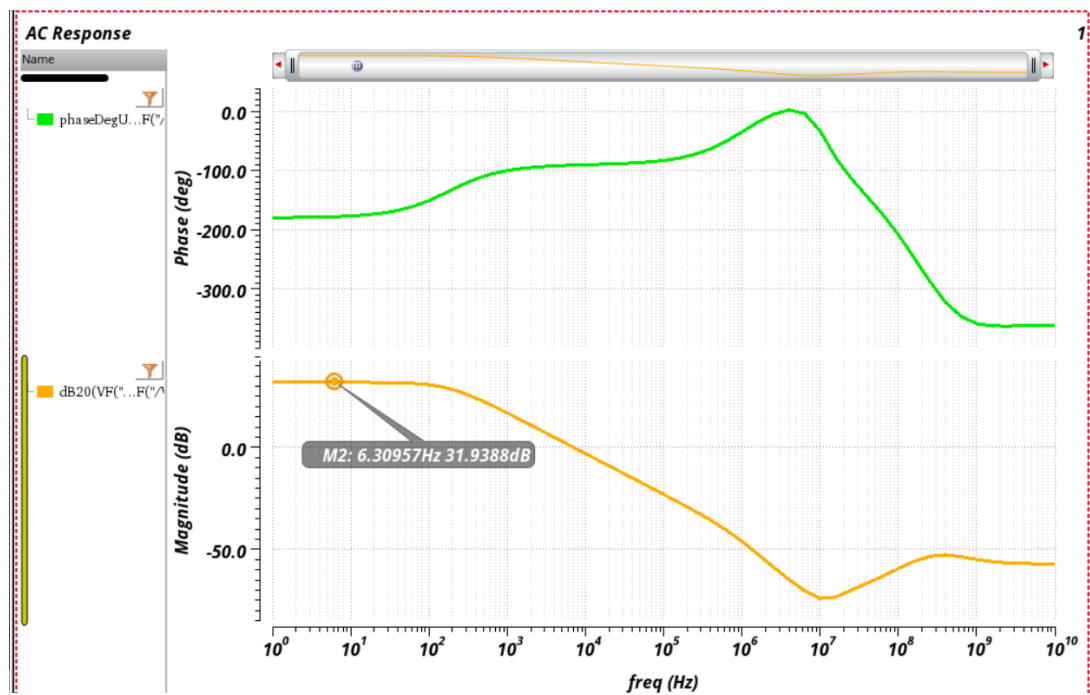


Figure 13: Magnitude plot for Common mode gain for nested miller compensation

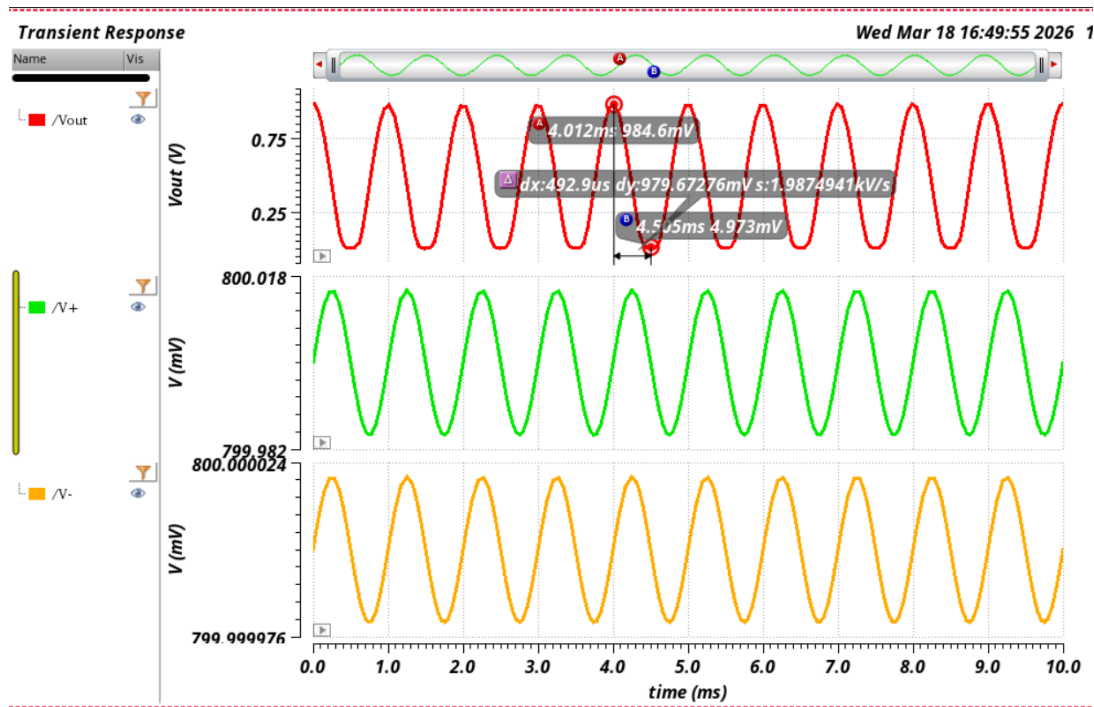


Figure 14: Output voltage swing for nested miller compensation

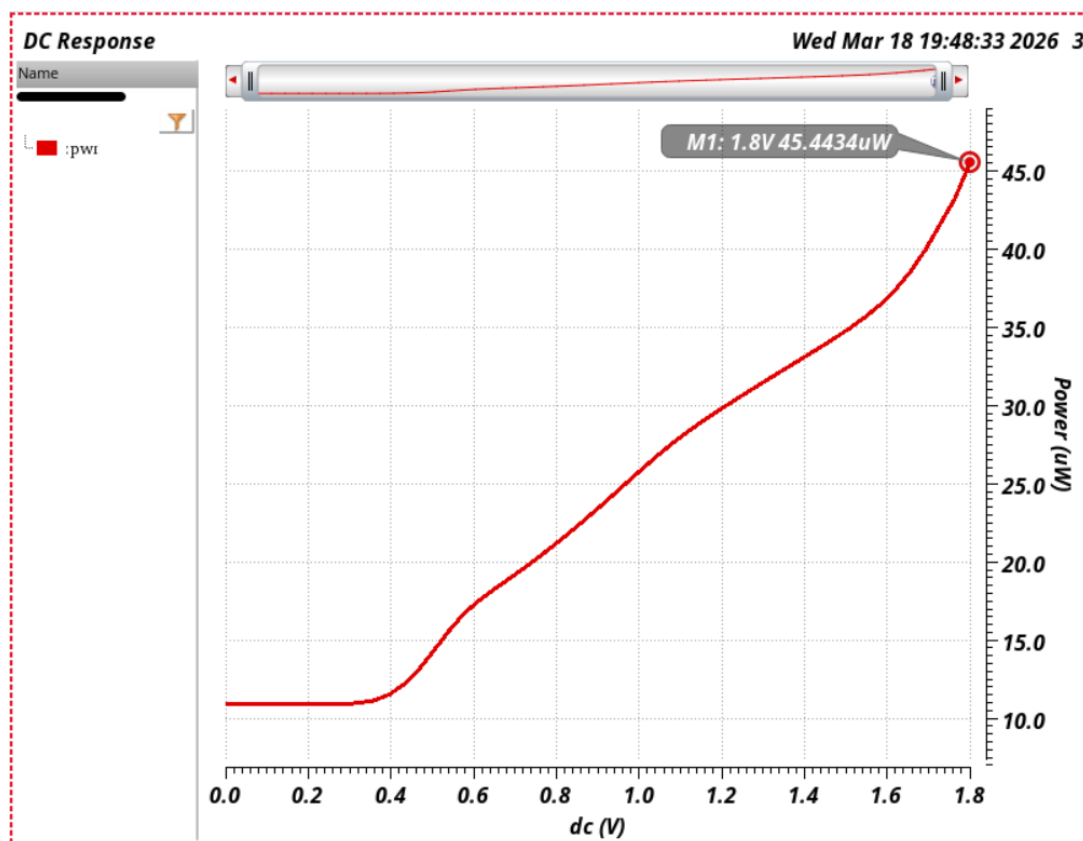


Figure 15: Power Consumption for nested miller compensation

2. Verify all the specifications of the designed Op-Amp over all the corners (tt, ff, ss, sf, fs).

Corner	Gain (dB)	CMRR (dB)	UGB (MHz)	PM (°)	Swing (V)
TT	152.604	120.6	14.0513	82.465	0.979
FF	145.6	95.6	11.364	11.8	0.893
SS					
SF	147.8	81	10.476	116.51	1.21
FS	157.6	93	13.15	40.8	1.1

Table 6: Performance of nested miller Op-Amp Across Process Corners

TASK 4: Comparison [3 pts]

Objective

Compare the Op-Amp designs implemented using Miller Compensation, Feedforward Compensation, and Nested Miller Compensation across all specified parameters and process corners. Comment on the stability of each circuit based on the observed results.

Comparison Based on Frequency Response Plots

The frequency response plots for Miller, Nested Miller, and Feedforward compensated Op-Amps are analyzed in terms of gain, unity gain bandwidth (UGB), and phase margin (PM).

1. Miller Compensation

The Miller compensated Op-Amp shows a unity gain bandwidth of approximately 10.12 MHz with a phase margin of around 52.3°. The gain is about 114 dB. The response is smooth with a clear dominant pole, indicating stable but bandwidth-limited behavior.

Observation: Stable system with moderate phase margin, but limited speed due to low UGB.

2. Nested Miller Compensation

The Nested Miller compensated Op-Amp achieves a very high gain of about 152.6 dB and a UGB of approximately 14.05 MHz. The phase margin is high ($\sim 82.4^\circ$ in this corner), indicating strong stability in this case.

3. Feedforward Compensation

The Feedforward compensated Op-Amp achieves the highest bandwidth with UGB around 56.7 MHz and gain of about 129 dB. The phase response shows phase margin (greater than 55°), ensuring stable operation.

Overall Comparison

- **Gain:** Nested Miller > Feedforward > Miller
- **Bandwidth (UGB):** Feedforward $\gg$ Nested Miller > Miller

Conclusion

Feedforward compensation provides the best overall performance with high bandwidth and reliable stability. Miller compensation is robust but slow, while Nested Miller achieves very high gain but suffers from stability issues across process variations.

Power Consumption Comparison

The DC power consumption plots for Miller, Feedforward, and Nested Miller compensated Op-Amps are compared.

- **Miller Compensation:** The power consumption reaches approximately $90.08 \mu W$ at 1.8 V.
- **Feedforward Compensation:** The power consumption is slightly higher, reaching around $93.64 \mu W$ at 1.8 V. This increase is due to additional circuitry used for feedforward paths, which improves bandwidth at the cost of higher power.
- **Nested Miller Compensation:** The Nested Miller design shows the lowest power consumption, approximately $45.44 \mu W$ at 1.8 V.

CMRR Comparison

The Common Mode Rejection Ratio (CMRR) of the three Op-Amp designs is compared as follows:

- **Miller Compensation:** Achieves a CMRR of approximately 120 dB, indicating good rejection of common-mode signals.
- **Feedforward Compensation:** Provides the highest CMRR of approximately 127 dB, demonstrating superior common-mode noise rejection.
- **Nested Miller Compensation:** Achieves a CMRR of approximately 120 dB, similar to Miller compensation, but may vary across process corners.

Observation

- Feedforward Compensation offers the best CMRR among the three designs.
- Miller and Nested Miller Compensation provide comparable CMRR performance.
- Higher CMRR in Feedforward design indicates improved differential operation and noise immunity.

Conclusion

Feedforward Compensation is the most effective in rejecting common-mode signals, while Miller and Nested Miller Compensation provide adequate but slightly lower CMRR performance.